PALOMA GUIRAL ORTUÑO

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Professional Summary

AI Software Engineer focused on backend and Generative AI, with hands-on experience building production APIs, microservices, LLM-powered systems and cloud infrastructure. Strong end-to-end ownership across architecture, data, deployment, observability and AI-assisted development.

Technical Skills

- Programming: Python, SQL, R, HTML, Matlab
- Generative AI & ML: LLMs, prompt engineering, OpenAI API, Amazon Bedrock, Claude, ElevenLabs, Ollama, LM Studio, NLP, embeddings and vector databases
- Backend & APIs: FastAPI, Flask, REST APIs, microservices, data extraction and automation services
- Cloud & DevOps: AWS (Lambda, EC2, ECS, S3, SageMaker, API Gateway, DynamoDB), Azure (Functions, CosmosDB, OpenAI), Docker, Terraform, Git, GitHub, CI/CD, MLOps, Linux
- Databases & Observability: PostgreSQL, Supabase, CosmosDB, DynamoDB, Neo4j, MariaDB/MySQL, Sentry

Professional Experience

AI Software Engineer - Backend & GenAI *April 2025 - Present*

Carina AI Studio | Spain

- Backend developer across AI-powered products, designing and implementing REST APIs, microservices and backend modules in scalable production environments.
- Design and refactoring of core product logic and architectures, contributing to feature proposals and large-scale restructuring of existing systems.
- Develop LLM-powered features for optimization, automation and structured data extraction using advanced prompting strategies, embeddings and vector-based retrieval patterns.
- Use AI-assisted software development and specialized coding agents to accelerate backend and occasional frontend workflows while retaining technical ownership of implementation and review.
- Create and maintain Terraform infrastructure, database schemas, migrations and backend integrations; support deployments and service integration on AWS.
- Work with Git/GitHub-based development workflows and CI/CD practices across product delivery.
- Monitor production reliability and triage exceptions through Sentry; participate in agile planning, sprint organization and cross-team technical discussions.

Data Scientist *Dec. 2023 - March 2025*

Radio Televisión Española | Madrid, Spain

- End-to-end development of a data-driven system to extract thematic insights from RTVE programming.
- Built ETL pipelines for ingestion, cleaning and integration of multi-department datasets; processed large volumes of VTT subtitle files into structured datasets.
- Applied LLMs and semantic clustering to segment and classify content across programming genres.

Back-end Developer *Feb. 2023 - Dec. 2023*

Bettermaps | Valencia, Spain

- Created APIs and chatbots using Azure Cognitive Search and Azure OpenAI, including prompt engineering and retrieval-oriented workflows.
- Managed databases and deployed backend solutions on AWS using Lambda, S3, API Gateway and DynamoDB.

Selected Independent Projects

Obrik - Founder & Product Engineer *April 2026 - Present*

Independent B2B SaaS project

- Conceived and built a B2B SaaS product end-to-end, taking ownership of product definition, backend architecture, implementation, design and launch.
- Built the backend with FastAPI and Supabase and delivered web/mobile product surfaces using AI-assisted development workflows.
- Owned technical and product decisions independently, including architecture, analytics, user acquisition experiments and go-to-market execution.

RISET - Co-founder & Developer

Mobile journaling & personal reflection app

- Co-designed and developed the mobile app end-to-end, from product concept and UX through architecture, business logic and data management, integrating AI-powered features for personalized journaling content.

Education

Master in Machine Learning *Sept. 2024*

Universidad Nacional de Educación a Distancia

Thesis: Artificial Neural Networks for stock price prediction

Bachelor's Degree in Data Science *June 2023*

Universidad Politécnica de Valencia (UPV)

Thesis: Application for monitoring operations in the stock market

Bachelor's Degree in Data Science - Exchange Program *Jan. 2023*

Technical University of Denmark (DTU)